BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

PHYSICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-C (Objective Type)

Time Allowed: 20 Minutes | **Maximum Marks:** 18

1- ہر سوال کے چار ممکنہ جوابات [illegible in source scan] 2- جوابات کی شیٹ (A، B، C، D) کے لیے نیلے یا کالے بال پوائنٹ کا استعمال کریں۔ 3- جواب میں ایک سے زیادہ دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. The unit of charge is:

(A) coulomb
(B) volt
(C) farad
(D) ampere.

Q2. Coulomb's force is:

(A) attractive
(B) repulsive
(C) both a,b
(D) none

Q3. The flux through spherical surface is:

(A) q/ε_0
(B) q^2/ε_0
(C) $2q/\varepsilon_0$
(D) $q/3\varepsilon_0$

Q4. The unit of current is:

(A) watt
(B) ampere
(C) volt
(D) joule

Q5. Relation between intensity and potential is:

(A) $E = -\Delta V/\Delta r$
(B) $V = \Delta E/\Delta r$
(C) $\Delta E/\Delta r$
(D) VE

Q6. 1eV =

(A) 1.6×10^{-19}J
(B) 1.27×10^{-7}J
(C) None of these
(D) $E^2\Delta V$

Q7. The relation for power dissipated is:

(A) P = IV
(B) $P = I^2R$
(C) $P = V^2/R$
(D) All of these

Q8. Magnetic field due to solenoid is given by,

(A) $B = \mu_o nI$
(B) $B = \mu_o n^2 I$
(C) $B = \mu_o^2 nI$
(D) $B = \mu_o nL$

Q9. The unit of conductance is:

(A) $mho.m^{-1}$
(B) mbo
(C) ohm
(D) ampere

Q10. For step up transformer

(A) $N_s > N_p$
(B) $N_s < N_p$
(C) $N_s = N_p$
(D) None

Q11. The resonance frequency is given by,

(A) $f_r = 1/2\pi\sqrt{LC}$
(B) $f_r = 1/4\pi\sqrt{LC}$
(C) $f_r = 1/\sqrt{LC}$
(D) $f_r = 2\pi/\sqrt{LC}$

Q12. The inductive reactance is given by:

(A) $X_L = 2\pi\sqrt{fL}$
(B) $X_L = 2\pi fL$
(C) $X_L = 1/2\pi fL$
(D) $X_L = 4\pi fL$

Q13. The unit of stress is:

(A) N/m
(B) N.m
(C) N/m^2
(D) Pascal.m

Q14. The strain energy is given by:

(A) $\frac{1}{2}F\ell$
(B) $F\ell$
(C) $\frac{1}{2}F^2\ell$
(D) $2F\ell$

Q15. Copper is:

(A) Paramagnetic
(B) Diamagnetic
(C) Ferromagnetic
(D) None

Q16. Rest mass of photon is:

(A) Zero
(B) 0.51kg
(C) 9.1×10^{-31}kg
(D) 1.67×10^{-27}kg

Q17. The De-Broglie wavelength is given by:

(A) $\lambda = hmv$
(B) $\lambda = h/mv$
(C) $\lambda = h/\sqrt{mv}$
(D) $\lambda = 2h/mc$

Q18. 1 curie =

(A) 3.7×10^{10}Bq
(B) 3.7×10^{-10}Bq
(C) 3.7×10^{8}Bq
(D) 3.7×10^{11}Bq

Section – B

Maximum Marks: 40

Note: Q-II Answer any TEN parts. Each part carries FOUR marks.

1. A heavy duty battery of a truck maintains a current of 3A for 24 hour. How much charge flows from the battery during this time?

2. What is the difference between an e.m.f and a P.D?

3. What is the difference between permeability and permittivity?

4. A charge particle moves in a straight line through a particular region of space. Could there a non-zero magnetic field in this region?

5. What is the relationship between P.D and P.E?

6. Can the potential of a non-uniformly charged sphere be the same as that of a point charge?

7. For a simple motor, why must the back e.m.f always be smaller than the applied P.D?

8. In Faraday's experiment, what would be the advantage of using coils with many turns?

9. What determines the gradient of a graph of inductive reactance against frequency?

10. If the peak value of a sine wave is 1000 volts, what is the effective value?

11. Does a high bulb at a temperature of 2500K produce as white light as the sun at 6000K?

12. Why do heavier nuclei have more neutrons than protons?

13. Discuss the properties of x-rays.

14. Explain fission chain reaction.

Section – C

Maximum Marks: 27

Note: Note : Attempt any THREE questions. All questions carry equal marks.

QIII. Q-III

(a) [5] State and explain coulomb's law of electrostatics.

(b) [4] Calculate the coulomb's force between two protons at a distance of 2m from each other.

QIV. Q-IV

(a) [5] Define and explain resistance and resistivity. On what factors the resistance of resistance and resistivity?

(b) [4] A 10 watt resistor has a value of 120Ω. What is the rated current through the resistor?

QV. Q-V

(a) [5] What is photoelectric effect? Explain Einstein's equation for photoelectric effect.

(b) [4] A metal whose work function is 3eV is illuminated by light of wavelength 3×10^{-7}m. Calculate the threshold frequency.

QVI. Q-VI

(a) [5] Explain natural radioactivity. Discuss the properties of α, β and ɣ-rays.

(b) [4] Explain stimulated emission and population inversion in case of laser.